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# Nitin Jain

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## EDUCATION

### PURDUE UNIVERSITY
MS in Computer Science
Machine Learning,
NLP and Optimization theory
Grad. May 2016 | West Lafayette, IN
Cum. GPA: 3.74

### R.V COLLEGE OF ENGINEERING
BEng in Computer Engineering
Grad. May 2013 | Bangalore, India
Conc. in Computer Engineering
Minor in Electrical Engineering
College of Engineering
Cum. GPA: 9.15 / 10

## LINKS
Github:// **ninja91**
LinkedIn:// **nitinajain**
Scholar:// **Google Scholar**
Twitter:// **@anitinjain**

## COURSEWORK

### GRADUATE
Numerical Linear Algebra
Statistical Machine Learning
Machine Learning Methods For NLP
Compilers & Programming Systems
Graph Theory
Computational Methods In Optimization
Bayesian Applied Decision Theory

### UNDERGRADUATE
Digital Signal Processing
Digital Communication
Linear Algebra ***(Research Asst.)***
Information Theory
Computer Architecture
Data Structures and Algorithms

## SKILLS

### PROGRAMMING
Over 5000 lines:
Python • C++ • Java
Shell • LaTeX
Over 1000 lines:
Matlab • JavaScript • Android • R •
Assembly
Familiar:
CSS • MySQL

## EXPERIENCE

### META | Machine Learning Engineer
Sep 2022 - Present | Seattle, WA
- Led the end-to-end ML lifecycle from EMG research to production model inference, culminating in the launch of the **Meta Neural Band** device; engineered training recipes, standardized evaluation suites, and hardware-emulator-in-the-loop benchmarking pipelines.
- Spearheaded research and production of on-device quantization, sparsity, compression, and power optimization for EMG and **handwriting models**; developed sub-8-bit/sparse models targeting strict power/latency budgets.
- Contributed quantization and inference optimizations to open-source **PyTorch** & **ExecuTorch** in mixed precision regimes (fp16/int8/int4) and for Attention, RNN modules.
- Collaborated with ARM to extend ExecuTorch, enabling quantization, lowering, and delegation of complex models to accelerators at peak performance.

### AMAZON.COM | Software Development Engineer, Alexa
June 2016 – Sep 2022 | Seattle, WA
- Led the development of a proactive, ML-driven latent goal-solving service scaling to 200+ TPS, doubling system throughput through lazy caching and rule engine optimizations.
- Launched an industry-first contextual understanding solution under Alexa's NLU service for multi-turn interactions, optimizing cloud model inference latency and compute footprint.

### CISCO SYSTEMS | Software Development Engineer
Aug 2013 – Aug 2014 | Bangalore, India
- Created a patentable Face detection based Auto-Focus algorithm with Nonlinear regression

## RESEARCH

### PURDUE UNIVERSITY | NLP Research | Prof. Goldwasser
Jan 2015 – May 2015 | West Lafayette, IN
Developed a hierarchical model to extract sports events and tag entities from Twitter's streaming API, curating a dataset of 1,000+ classified tweets.

### PURDUE UNIVERSITY | Optimization & Matrix computations Research
Aug 2015 – May 2016 | West Lafayette, IN
Researched kernel learning algorithms over convex spaces (Prof. Honorio) and designed memory-efficient structures to extract a 150GB DNS census bipartite graph (Prof. Gleich).

### R.V COLLEGE OF ENGINEERING | Undergraduate Researcher
Jun 2012 – May 2013 | Bangalore, India
Researched fast numerical methods to compute the $n^{th}$ root of a number.

## PATENT

### USER-SYSTEM DIALOG EXPANSION | U.S. Patent 11,527,237
Granted Dec 2022 | Patent Link
Recommending skill experiences to a user after a dialog session has concluded using machine learning models to identify and rank candidate intents.

## PUBLICATIONS

[1] N. Jain, K. Murthy, and Hamsapriye. Matrix methods for finding nth root of m. *IJMEST*, 2014.
[2] N. Jain, K. Murthy, and Hamsapriye. On finding nth root of m leading to newton raphson's improved method. *IJETCAS*, 2014.